

H.W-3

1) Using Naive Bayes, we know all features are conditionally independent,

Step 1, $P(\text{sold} = \text{yes}) = \frac{6}{11}$

$P(\text{sold} = \text{No}) = \frac{5}{11}$

Now,

Brand	sold Yes	sold No
A	2/6	2/5
B	2/6	2/5
C	2/6	1/5

$$P(x|y) = P(x_1|y) \\ P(x_2|y) \\ P(x_3|y)$$

Now,

Price Range	Yes	No
Low	2/6	2/5
Medium	2/6	1/5
High	2/6	2/5

Condition	Yes	No
New	3/6	2/5
Used	3/6	3/5

Now, Step 2, $\uparrow B$ $\uparrow$ Medium $\uparrow$ Used

$$P(\text{sold} = \text{Yes} | \text{Brand}, \text{Price Range}, \text{Condition}) \\ = P(\text{Brand} | \text{sold} = \text{Yes}) P(\text{Price Range} | \text{sold} = \text{Yes}) P(\text{Condition} | \text{sold} = \text{Yes}) \\ = \frac{2}{6} \times \frac{2}{6} \times \frac{3}{6} = 0.05555$$

Similarly,

$$P(\text{sold} = \text{No} | \text{Brand} = B, \text{Price Range} = \text{Medium}, \text{Condition} = \text{Used})$$

$$= P(\text{Brand} = B \mid s = \text{No}) \cdot P(\text{Price Range} = \text{Medium} \mid s = \text{No}) \\ P(\text{Condition} = \text{Used} \mid s = \text{No})$$

$$= \frac{2}{5} \times \frac{1}{5} \times \frac{3}{5} = 0.048$$

$$P(\text{sold} = \text{yes} \mid x) > P(\text{sold} = \text{no} \mid x)$$

so, it will sell, Probability = $\frac{0.0555}{0.0555 + 0.048} \times 100$

yes it will sell

$$P = 0.53648$$

$$= 0.53648 \times 100$$

$$= 53.648 \%$$

chance

$$2) \triangleright \text{Entropy}(S) = - \sum_{i=1}^C P_i \log(P_i)$$

Here, $P_M = \frac{5}{10}$

$$P_H = \frac{5}{10}$$

$$\text{So, } H(S) = - \frac{5}{10} \log_2\left(\frac{5}{10}\right) - \frac{5}{10} \log_2\left(\frac{5}{10}\right) = 1$$

Entropy for Green,

No $\rightarrow$ $\begin{cases} 2 \text{ M} \\ 4 \text{ H} \end{cases}$ (No Green)

$$\text{Entropy} = -\log_2\left(\frac{2}{6}\right) \times \frac{2}{6} - \log_2\left(\frac{4}{6}\right) \times \frac{4}{6} = 0.918296$$

Yes $\rightarrow$ $\begin{cases} 3 \text{ M} \\ 1 \text{ H} \end{cases}$

$$\text{Entropy} = -\log_2\left(\frac{3}{4}\right) \times \frac{3}{4} - \log_2\left(\frac{1}{4}\right) \times \frac{1}{4} = 0.811278$$

$$I_{\text{Green}} = 1 - \left(\frac{4}{10} \times 0.9113 + \frac{6}{10} \times 0.8918296 \right)$$

$$= 1 - 0.8754976 = 0.124502$$

For, legs,

$$2 \rightarrow H(l=2) = -\log_2\left(\frac{5}{7}\right) \times \frac{5}{7} - \log_2\left(\frac{2}{7}\right) \times \frac{2}{7} = 0.86$$

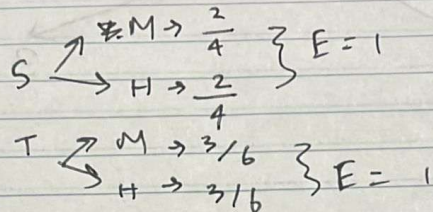
$$3 \rightarrow H(l=3) = -\log_2\left(\frac{3}{10}\right) \times \frac{3}{10} - \log_2\left(\frac{7}{10}\right) \times \frac{7}{10} = 0$$

Now

$$I_G = 1 - \left(\frac{3}{10} \times 0 + \frac{7}{10} \times 0.8631 \right)$$

$$IG_{legs} = 0.3958$$

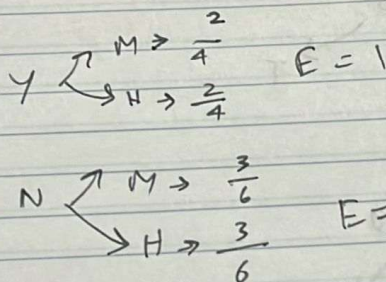
Height,



$$IG_{height} = 1 - \left(1 \times \frac{4}{10} + 1 \times \frac{6}{10} \right)$$

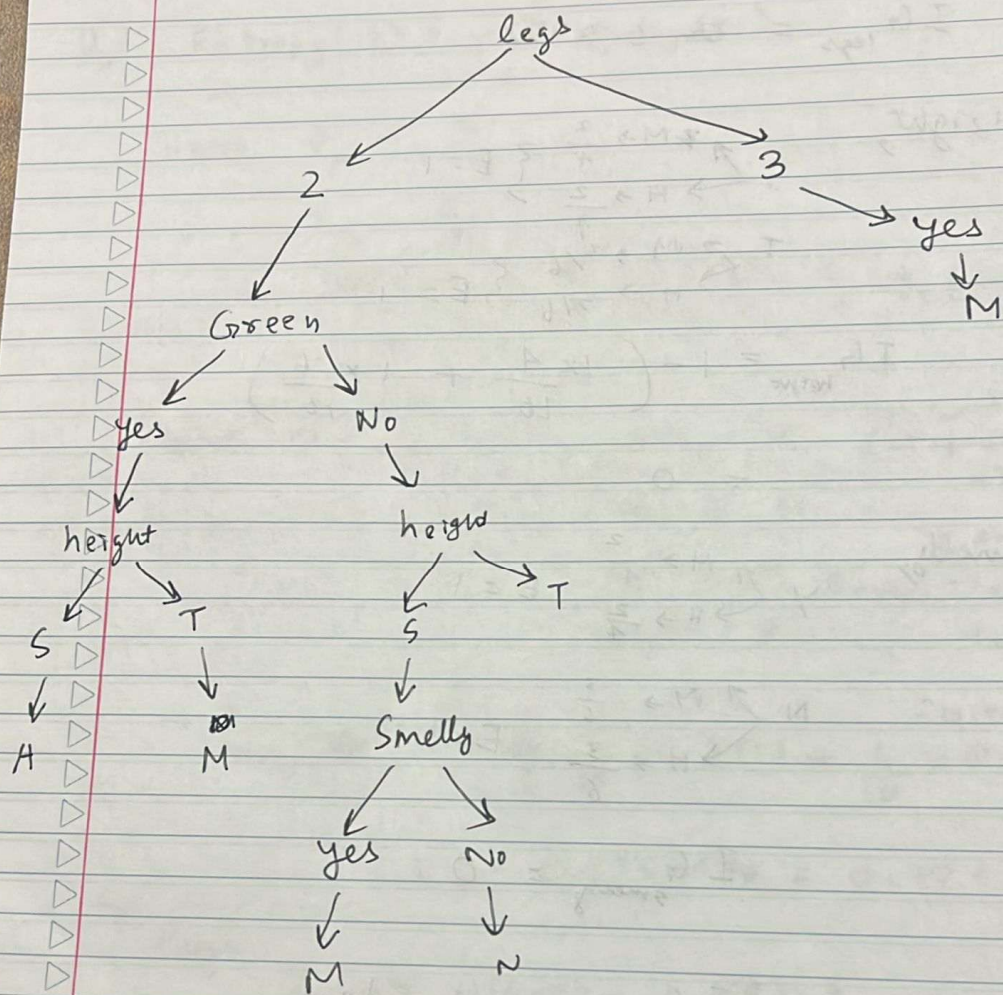
$$= 0$$

Smelly,



$$IG_{smelly} = 0$$

Now, highest IG is from legs,



Q3) we trained 8 SVR models using both linear and RBF kernels with different values of C . The performance was compared using RMSE and MAPE.

From the results, the **linear models performed better than the RBF models**. In particular, **SVR4 (Linear, $C = 10$)** achieved the lowest RMSE and MAPE, making it the best model.

On the other hand, all RBF models had higher RMSE and MAPE, indicating poorer performance. This suggests that the relationship in the dataset is mostly linear, so the linear kernel was more suitable.

Q4) In Task 1, using $K = 3$, the model achieved a test accuracy of **93.33%**. The confusion matrix showed that all samples of setosa and versicolor were classified correctly, while a few virginica samples were misclassified as versicolor.

In Task 2, we tested K values from 1 to 15 using the validation set. The highest validation accuracy (~96.67%) was achieved at $K = 1$. After retraining with this value on the combined training and validation data, the test accuracy was again **93.33%**.

Overall, both tasks produced similar performance. However, noting that $K=1$ might cause overfitting.